

Integrating Zerodha Option Trading Costs into Nautilus Trader

This document maps out what cost components are **natively supported**, **customizable**, or **manually required** when implementing Zerodha-style options trading in Nautilus Trader.

Built-in / Natively Supported in Nautilus

Component	Nautilus Support?	Notes
Custom Fee Models	Yes	Subclass <code>ExecutionCostModel</code> ; supports % or fixed fees per order
Slippage Models	Yes	Subclass <code>SlippageModel</code> ; simulate fixed or dynamic slippage
Position-level PnL	Yes	Net of entry/exit price and quantity
Fill-level simulation	Yes	Handles partial fills and timestamps
Currency support	Yes	Use <code>Price(currency="INR")</code> to stay compliant



Customizable via Nautilus Hooks

Component	Requires Custom Code?	Notes
Flat ₹20/order	Yes	Implement via custom <code>ExecutionCostModel</code>
Percentage-based premium fee	Yes	Needed for STT, exchange charges, etc.
Per-leg tax differences	Yes	STT on sell-side, stamp duty on buy-side
Premium-based turnover	Yes	You must calculate on premium, not notional value
Order-type-specific charges	Yes	Zerodha may apply charges for market orders (e.g. slippage/spread)

Not Natively Modeled - Must Be Calculated Separately

Component	Must Be External?	Notes
GST (18% on fees)	Yes	Not included in fee model; best to post-process
SEBI turnover charges	Yes	Small but must be added in reporting
Stamp Duty (Buy side only)	Yes	State-specific, calculated externally
Exchange transaction charge	Yes	Typically % of premium turnover; add manually
Regulatory compliance check	Yes	Like checking max OI, position limits - out of scope for Nautilus core

Summary Recommendations

Area	What to Use	----- -----	Brokerage Fee
ExecutionCostModel (₹20/order)		Slippage	SlippageModel (fixed/spread %)
	STT, Stamp, GST, SEBI	Custom fee logic or external reporting script	
Premium-based Charges	Calculate manually using fill-level premium data		Final PnL with Deductions
Adjust PnL after simulation using full cost breakdown			